

Academic Excellence - Personal Best - Giving Back to the Community

Semester 2 Report 2022

Logan CAMPBELL

Year 12

List of Subjects

English Advanced

Chemistry

Engineering Studies

Mathematics Extension 1

Mathematics Extension 2

Physics



Year Adviser: Richard Crooks



Principal: Kathryn O'Sullivan



2022 Semester 2 Report - Year 12

Level of Achievement

Achievement Achievement Description

Outstanding	The student has an extensive knowledge and understanding of the content and can readily apply this knowledge. In addition, the student has achieved very high levels of competence in the processes and skills and can apply these skills to new situations.
High	The student has a thorough knowledge and understanding of the content and a high level of competence in the processes and skills and can apply these skills to most situations.
Sound	The student has a sound knowledge and understanding of the main areas of content and has achieved an adequate level of competence in the processes and skills.
Basic	The student has a basic knowledge and understanding of the main areas of content and has achieved a limited level of competence in the processes and skills.
Limited	The student has an elementary knowledge and understanding in a few areas of content and has achieved a very limited level of competence in some of the processes and skills.

NBSC Manly Campus can provide written information, following parent application, showing your child's achievement compared to his or her peer group at our school. This information shows the number of students in each of the five achievement levels based on the "Overall Level of Achievement" for each subject.

Additional Programs and Activities

Sydney North Athletics Championships

Sydney North Cross Country Championships

Warringah Zone Athletics Championships

Warringah Zone Cross Country Championships

Attendance (Semester 2)

Whole Days Absent 0



2022 Semester 2 Report - Year 12

Subject & Profile Description		Level Of Achievement			
		Consistently	Often	Sometimes	Rarely
English Advanced	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Chemistry	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Engineering Studies	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Mathematics Extension 1	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			

Subject & Profile Description		Level Of Achievement			
		Consistently	Often	Sometimes	Rarely
Mathematics Extension 2	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Physics	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			



2022 Semester 2 Report - Year 12

English Advanced

Name: Logan CAMPBELL

Class: 12ENAA1

Teacher: Rachele Haigh

Comment on Progress

Logan has worked diligently in order to achieve outstanding results, reflecting his nuanced understanding of the skills and concepts required by the course. Logan's determination to succeed is evidenced in the way that he consistently seeks and applies feedback in order to improve his written compositions. With continued revision and refining of his written responses, Logan has the capacity to achieve fine results in the Higher School Certificate. He is to be commended for his hard work and positive attitude.

Learning Outcomes

Outcomes	O	H	S	B	L
A student independently responds to, composes and evaluates a range of complex texts for understanding, interpretation, critical analysis, imaginative expression and pleasure.	✓				
A student uses, evaluates and justifies processes, skills and knowledge required to effectively respond to and compose texts in different modes, media and technologies.	✓				
A student thinks imaginatively, creatively, interpretively, critically and discerningly to respond to, evaluate and compose texts that synthesise complex information, ideas and arguments.	✓				
A student investigates and evaluates the relationships between texts.	✓				
A student evaluates the diverse ways texts can represent personal and public worlds and recognises how they are valued.		✓			

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2022 Semester 2 Report - Year 12

Chemistry

Name: Logan CAMPBELL

Class: 12CHEMA6

Teacher: Kevin Collins

Comment on Progress

Logan is a very capable and diligent student who is to be congratulated for his outstanding results. He has consistently demonstrated a comprehensive understanding of the course content and is able to apply this knowledge to solve complex problems. Logan has a high level of competence in a wide range of scientific skills including secondary research, communication and first-hand, practical investigation. He takes full advantage of teacher assistance and participates in all types of learning activities, while his contributions in class are highly valued due to his willingness to work collaboratively.

Learning Outcomes

Outcomes	O	H	S	B	L
Analyses and evaluates primary and secondary data and information.	✓				
Solves scientific problems using primary and secondary data, critical thinking skills and scientific processes.	✓				
Describes and evaluates chemical systems used to design and analyse chemical processes.	✓				
Analyses the structure of, and predicts reactions involving carbon compounds.	✓				
Explains the characteristics of equilibrium systems, and the factors that affect these systems	✓				
Describes, explains and quantitatively analyses acids and bases using contemporary models	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2022 Semester 2 Report - Year 12

Engineering Studies

Name: Logan CAMPBELL

Class: 12ENGSTA5

Teacher: Joanne Drew

Comment on Progress

Logan has demonstrated outstanding analytical skills this semester and is commended both for his academic approach to studying engineering and his practical application of material knowledge and construction techniques. He designed and constructed a high quality quadcopter propeller and submitted an excellent research report in this practical assessment task. Logan's final assessment mark was impressive and it is recommended he continue to practice examination style questions under timed conditions for the Higher School Certificate Examination. Logan has shown himself to be an intelligent, collaborative engineering student and he is wished well for his future endeavours.

Learning Outcomes

Outcomes	O	H	S	B	L
Justifies the selection of materials and components in engineering.	✓				
Analyses engineering applications in specific fields.	✓				
Demonstrates proficiency in the use of mathematical, scientific and graphical methods of analysis.	✓				
Demonstrates an understanding of engineering drawing and drawing standards.	✓				
Demonstrates skill in research and problem solving related to engineering.	✓				
Uses appropriate written, oral and presentation skills in the preparation of detailed engineering reports.	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2022 Semester 2 Report - Year 12

Mathematics Extension 1

Name: Logan CAMPBELL

Class: 12MAX1A2

Teacher: Kerrienne Lee

Comment on Progress

Logan has been an exceptionally diligent student throughout this course. He shows great maturity and focus when it comes to learning and refining his mathematical problem solving skills. As a result of his hard work, Logan has grown as a mathematical thinker and is highly capable of completing problems independently. He is also highly capable of explaining the concepts to his peers.

Learning Outcomes

Outcomes	O	H	S	B	L
Applies advanced concepts and techniques in simplifying expressions involving compound angles and solving trigonometric equations.	✓				
Uses calculus in the solution of applied problems, including differential equations and volumes of solids of revolution.	✓				
Applies appropriate statistical processes to present, analyse and interpret data.	✓				
Uses calculus in the solution of applied problems, including integration and inverse trigonometry.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2022 Semester 2 Report - Year 12

Mathematics Extension 2

Name: Logan CAMPBELL

Class: 12MAX2A8

Teacher: Robert Harris

Comment on Progress

Logan is to be commended on his approach to this course. Although he is a very capable mathematician, he is also very aware that he must consolidate his knowledge through significant practice if he is to achieve his potential. Logan is very active in ensuring he understands the topics presented and is to be congratulated on his outstanding results. Given his approach, he should expect to see this level of success continue in his final HSC examinations.

Learning Outcomes

Outcomes	O	H	S	B	L
Chooses appropriate strategies to construct arguments and proofs in both practical and abstract settings.	✓				
Uses vectors to model and solve problems in two and three dimensions.	✓				
Uses the relationship between algebraic and geometric representations of complex numbers and complex number techniques to prove results, model and solve problems.		✓			
Applies techniques of integration to structured and unstructured problems		✓			

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2022 Semester 2 Report - Year 12

Physics

Name: Logan CAMPBELL

Class: 12PHYSC4

Teacher: Richard Crooks

Comment on Progress

Logan is a very capable and motivated student whose determination to succeed is reflected in the outstanding results that he attained this semester. He works diligently to extend his knowledge and understanding of key concepts in Physics and participates effectively in both independent tasks and team activities. He carefully analyses and compiles extensive and detailed notes to then communicate information succinctly and logically using appropriate scientific formats. Logan always strives to achieve his personal best and is to be congratulated on these excellent outcomes.

Learning Outcomes

Outcomes	O	H	S	B	L
Processes, analyses and evaluates qualitative and quantitative data and information, from both primary and secondary sources, using a range of appropriate media.	✓				
Solves scientific problems using primary and secondary data, critical thinking skills and scientific processes.	✓				
Communicates scientific understanding using suitable language and terminology for a specific audience or purpose.	✓				
Describes, explains and analyses qualitatively and quantitatively, circular, gravitational, projectile motion and electric and magnetic interactions.	✓				
Describes, explains and analyses evidence for the properties of light, and evaluates the implications of this for modern theories of physics including astronomical events.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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Academic Excellence - Personal Best - Giving Back to the Community

Semester 2 Report 2021

Logan CAMPBELL

Year 11

List of Subjects

English Advanced

Chemistry

Engineering Studies

HSC Mathematics Advanced

Mathematics Extension 1

Personal Development, Health and Physical Education

Physics



Year Adviser: Richard Crooks



Principal: Kathryn O'Sullivan



2021 Semester 2 Report - Year 11

Level of Achievement

Achievement Achievement Description

Outstanding	The student has an extensive knowledge and understanding of the content and can readily apply this knowledge. In addition, the student has achieved very high levels of competence in the processes and skills and can apply these skills to new situations.
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Basic	The student has a basic knowledge and understanding of the main areas of content and has achieved a limited level of competence in the processes and skills.
Limited	The student has an elementary knowledge and understanding in a few areas of content and has achieved a very limited level of competence in some of the processes and skills.

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Additional Programs and Activities

Student Wellbeing Advocate	Sydney North Athletics Championships Monday 19/7/21
Sydney North Athletics Championships Tuesday 20/7/21	Sydney North Cross Country Championships 4/6/21
Warringah Zone Athletics Championships 21/5/21	

Attendance (Semester 2)

Whole Days Absent	0	Explained Whole Days Absent	0
Unexplained Whole Days Absent	0		



2021 Semester 2 Report - Year 11

Subject & Profile Description		Level Of Achievement			
		Consistently	Often	Sometimes	Rarely
English Advanced	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Chemistry	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Engineering Studies	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
HSC Mathematics Advanced	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			

Subject & Profile Description		Level Of Achievement			
		Consistently	Often	Sometimes	Rarely
Mathematics Extension 1	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Personal Development, Health and Physical Education	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			
Physics	Applies creative and innovative thinking to learning.	✓			
	Collaborates and communicates effectively.	✓			
	Demonstrates critical thinking about information, ideas and problems.	✓			
	Demonstrates organisational and time-management skills.	✓			
	Monitors own learning and applies feedback to achieve personal best.	✓			



2021 Semester 2 Report - Year 11

English Advanced

Name: Logan CAMPBELL

Class: 11ENAA1

Teacher: Rachele Haigh

Comment on Progress

Logan is a motivated student who always strives to achieve his personal best by seeking and applying feedback. His written expression is exceptional, his textual analysis is considered and informed and his creative compositions are remarkably engaging. To improve, Logan should practise writing responses under timed conditions. He is to be commended for his hard work and positive attitude.

Learning Outcomes

Outcomes	O	H	S	B	L
A student analyses and uses language forms, features and structures of texts considering appropriateness for specific purposes, audiences and contexts and evaluates their effects on meaning.		✓			
A student thinks imaginatively, creatively, interpretively and critically to respond to, evaluate and compose texts that synthesise complex information, ideas and arguments.		✓			
A student reflects on, evaluates and monitors own learning and adjusts individual and collaborative processes to develop as an independent learner.	✓				
A student responds to, composes and evaluates complex texts for understanding, interpretation, critical analysis, imaginative expression and pleasure.	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 2 Report - Year 11

Chemistry

Name: Logan CAMPBELL

Class: 11CHEMA6

Teacher: Anna Molnar

Comment on Progress

Logan is a committed and conscientious student who has made excellent progress in Chemistry this semester. He has demonstrated extensive knowledge and understanding across all of the chemical concepts studied, as evidenced by his consistently outstanding grades. He successfully adapted to learning from home and has continued to engage with all online activities. Logan is encouraged to maintain his positive attitude and mature approach to his studies for further success in Year 12.

Learning Outcomes

Outcomes	O	H	S	B	L
Designs and evaluates investigations in order to obtain primary and secondary data and information.	✓				
Communicates scientific understanding using suitable language and terminology for a specific audience or purpose.	✓				
Describes, applies and quantitatively analyses the mole concept and stoichiometric relationships.	✓				
Explores the many different types of chemical reactions, in particular the reactivity of metals, and the factors that affect the rate of chemical reactions.	✓				
Analyses the energy considerations in the driving force for chemical reactions.	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 2 Report - Year 11

Engineering Studies

Name: Logan CAMPBELL

Class: 11ENGSTA5

Teacher: Joanne Drew

Comment on Progress

Logan produced an outstanding report on the design and construction of a biomedical arm, demonstrating an excellent ability to research and problem solve engineering applications. He is engaged and actively participates in class, whether in person or during remote learning. This application to the subject resulted in an impressive Year 11 Examination result, which demonstrated his analytical ability, knowledge of engineering applications and understanding of the societal impacts of engineering. Overall, Logan is to be commended for his consistently outstanding effort and dedication to learning in Engineering Studies.

Learning Outcomes

Outcomes	O	H	S	B	L
Justifies the selection of materials and components in engineering.	✓				
Analysis engineering applications in specific fields.	✓				
Demonstrates proficiency in the use of mathematical, scientific and graphical methods of analysis.	✓				
Demonstrates an understanding of engineering drawing and drawing standards.	✓				
Demonstrates skill in research and problem solving related to engineering.	✓				
Uses appropriate written, oral and presentation skills in the preparation of detailed engineering reports.	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 2 Report - Year 11

HSC Mathematics Advanced

Name: Logan CAMPBELL

Class: 11MAHA2

Teacher: Kerrienne Lee

Comment on Progress

Logan has demonstrated a capacity to work independently with success. He has committed himself to learning the content of the course and solving a range of problems with minimal guidance. As a result of his efforts, Logan has developed a thorough understanding of the course content and is able to successfully complete problems in class and in assessments.

Learning Outcomes

Outcomes	O	H	S	B	L
Applies the concepts and techniques of periodic functions in the solution of problems involving trigonometric graphs.	✓				
Solves problems using appropriate statistical processes.	✓				
Constructs arguments to prove and justify results and provides reasoning to support conclusions which are appropriate to the context.	✓				
Applies calculus techniques to model and solve problems and applies the concepts and techniques of arithmetic and geometric sequences and series in the solution of problems.	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 2 Report - Year 11

Mathematics Extension 1

Name: Logan CAMPBELL

Class: 11MAX1B8

Teacher: Mark Harrison

Comment on Progress

Logan has shown a considerable amount of focus and determination in this course and is making excellent progress. He shows commitment each lesson, to learning the skills that are covered and developing efficiency with applying them in solving problems of various types. Logan is encouraged to maintain his application and dedication to his work so that he continues to achieve outstanding results next year.

Learning Outcomes

Outcomes	O	H	S	B	L
Manipulates algebraic expressions and graphical functions to solve problems.	✓				
Uses appropriate technology to investigate, organise and interpret information to solve problems in a range of contexts.	✓				
Uses algebraic and graphical concepts in the modelling and solving of problems involving functions and their inverses, applies concepts and techniques of inverse trigonometric functions and simplifying expressions involving compound angles in the solution of problems.	✓				
Applies understanding of the concept of a derivative in the solution of problems, including rates of change, exponential growth and decay and related rates of change and uses concepts of permutations and combinations to solve problems involving counting or ordering.	✓				

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Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 2 Report - Year 11

Personal Development, Health and Physical Education

Name: Logan CAMPBELL

Class: 11PDHPEA3

Teacher: Jennifer Walker

Comment on Progress

Logan is a dedicated student who has achieved excellent results this semester. He has demonstrated an extensive knowledge and understanding of course content and can readily apply his knowledge to a variety of situations. He regularly asks questions to further clarify his understanding and shares his knowledge and experience with others in the class.

Learning Outcomes

Outcomes	O	H	S	B	L
Identifies and examines why individuals give different meanings to health	✓				
Proposes actions that can improve and maintain an individual's health		✓			
Plans for participation in physical activity to satisfy a range of individual needs.	✓				
Demonstrates strategies for the assessment, management and prevention of injuries in first aid settings (Option 1)	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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2021 Semester 2 Report - Year 11

Physics

Name: Logan CAMPBELL

Class: 11PHYSC4

Teacher: Richard Crooks

Comment on Progress

Logan is a very capable student who has worked consistently and conscientiously throughout the semester. His result in the Yearly Examination was excellent while his Depth Study demonstrated that he has extensive knowledge and can apply it to unfamiliar situations using correct scientific formats. Logan's thoughtful approach makes him an effective leader in his team for first-hand investigations. He actively participates in class discussions where his insightful comments and questions reflect his level of engagement and understanding of concepts. He is a motivated and independent worker and with regular revision, Logan should continue to achieve outcomes commensurate with his ability.

Learning Outcomes

Outcomes	O	H	S	B	L
Analyses and evaluates primary and secondary data and information.	✓				
Communicates scientific understanding of thermodynamics using suitable language and terminology for a specific audience or purpose.	✓				
Describes and explains events in terms of Newton's Laws of Motion, the law of conservation of momentum and the law of conservation of energy.	✓				
Solves scientific problems using Laws of Physics and critical thinking skills.	✓				
Explains and quantitatively analyses waves, electric fields, circuitry and magnetism.	✓				

O = Outstanding, H = High, S = Sound, B = Basic, L = Limited

Overall Achievement

Outstanding	High	Sound	Basic	Limited
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